MATH 152 **QUIZ 7** NAME _____

Mrs. Bonny Tighe 25 points

11.8 TA _____Wed 4/10/2024

1. Expand $\sum_{n=0}^{\infty} \frac{(-1)^n (x+2)^n}{(1 \cdot 3 \cdot 5 \cdots (2n+1)}$ as a polynomial to the fifth degree.

2. Find the radius of convergence and interval of convergence of the power series.

a) $\sum_{n=0}^{\infty} \frac{2^n (x-2)^n}{(2n)!}$

b) $\sum_{n=0}^{\infty} \frac{(-1)^n \ (x+1)^{2n}}{3n+4}$

c) $\sum_{k=0}^{\infty} \frac{(-1)^k (1 \cdot 4 \cdot 7 \cdots (3k+1))(2x-3)^{k+1}}{k^3+1}$

d) $$\sum_{n=2}^{\infty} \frac{(-1)^{n+1}(x-1)^{2n}}{2^{n+1}(n^3+2)}$$

MATH 152 **QUIZ 7A** NAME _____

Mrs. Bonny Tighe 25 points

11.8 TA _____Wed 4/10/2024

1. Expand $\sum_{n=0}^{\infty} \frac{(-1)^n (x+2)^n}{2^n (2n)!}$ as a polynomial to the fourth degree.

2. Find the radius of convergence and interval of convergence of the power series.

a) $\sum_{n=0}^{\infty} \frac{(x+3)^n}{2^n (n+2)}$

b) $\sum_{n=0}^{\infty} \frac{(-1)^n (1 \cdot 4 \cdot 7 \cdots (3n+1))(x-3)^n}{5^n}$

c) $\sum_{n=0}^{\infty} \frac{(x+1)^{2n}}{4^n (n^3+2)}$

d) $\sum_{n=0}^{\infty} (-1)^n \frac{(2x+1)^{n+1}}{3^n \; n!}$